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Education	Ph.D in Statistics , University of Geneva, Switzerland	08/2015-07/2021
	- Thesis: Contributions to time series analysis - Co-advisors: Prof. Maria-Pia Victoria-Feser, Prof. Stéphane Guerrier	
	M.Sc in Statistics , University of Illinois at Urbana-Champaign, USA	01/2014-05/2015
	M.Sc in Applied Statistics , Dongbei University of Finance and Economics, China	09/2011-07/2013
	Bachelor in Statistics , Anhui University of Finance and Economics, China	09/2007-07/2011
Academic Positions	Assistant Professor , Auburn University, USA	08/2025-present
	Assistant Professor (Harrison Early Career) , University of Warwick, UK	09/2023-07/2025
	Postdoctoral Researcher , Pennsylvania State University, USA	10/2022-08/2023
	- SNSF Postdoc.Mobility Fellowship - Hosted by Prof. Runze Li	
	Postdoctoral Researcher , Université Catholique de Louvain, Belgium	06/2022-09/2022
	- SNSF Postdoc.Mobility Fellowship - Hosted by Prof. Johan Segers	
	Postdoctoral Researcher , University of Warwick, UK	08/2021-05/2022
	Advisor: Prof. Yi Yu	
Research interests	- Nonparametric estimation in high-dimensions - Change point detection and inference - High-dimensional inference under temporal dependence and contamination	
Publications	Xu, H. , Luo, D., Guerrier, S., Li, R., & Ke, Y. “ <i>Non-asymptotic analysis of median-of-means estimation for high-dimensional time series</i> ”, Statistical Learning and Data Science, 100008, 2026. Full text: https://doi.org/10.1016/j.slads.2026.100008 .	
	Luo, X., Xu, H. , Padilla, C.M.M., & Padilla, O.H.M. “ <i>Online Change Point Detection for Multivariate Inhomogeneous Poisson Processes Time Series</i> ”, Forty-Third International Conference on Machine Learning (ICML), 2026. Full text: https://arxiv.org/abs/2601.20192 .	
	Xue, G., Xu, H. , & Yu, Y., “ <i>Change point localisation and inference in fragmented functional data</i> ”, Bernoulli, 32(2): 1456-1480, 2026. Full text: https://doi.org/10.3150/25-BEJ1914 .	
	Kumar, S., Xu, H. , Cho, H., & Wang, D., “ <i>Estimation and inference for change points in functional regression time series</i> ”, Statistica Sinica (to appear), 2026. Full text: https://arxiv.org/abs/2405.05459 .	

Xu, H., Dubey, P., & Yu, Y., “*Online network change point detection with missing values and temporal dependence*”, Journal of Time Series Analysis, 47(3), 687-700, 2026.
Full text: <https://doi.org/10.1111/jtsa.70023>.

Xu, H., Wang, D., Zhao, Z., & Yu, Y., “*Change point inference in high-dimensional regression models under temporal dependence*”, Annals of Statistics, 52(3) 999-1026, 2024.
Full text: <https://doi.org/10.1214/24-AOS2380>.

Padilla, C.M.M., **Xu, H.**, Wang, D., Padilla, O.H.M., & Yu, Y., “*Change point detection and inference in multivariable nonparametric models under mixing conditions*”, NeurIPS, 2023. Full text: [Link](#).

Yu, Y., Chatterjee S., & **Xu, H.**, “*Localising change points in piecewise polynomials of general degrees*”, Electronic Journal of Statistics, 16(1), 1855-1890, 2022.
Full text: <https://doi.org/10.1214/21-EJS1963>.

Guerrier, S., Molinari, R., Victoria-Feser, M. P., & **Xu, H.**, “*Robust two-step wavelet-based inference for time series models*”, Journal of the American Statistical Association, 117(540), 1996-2013, 2022. (alphabetical order). Full text: <https://doi.org/10.1080/01621459.2021.1895176>.

Guerrier, S., Jurado, J., Khaghani, M., Bakalli, G., Karemera, M., Molinari, R., Orso, S., Raquet, J., Kabban, C.M.S., Skaloud, J., **Xu, H.**, & Zhang, Y., “*Wavelet-based moment-matching techniques for inertial sensor calibration*”, IEEE Transactions on Instrumentation and Measurement, 69(10), 7542-7551, 2020. Full text: <https://doi.org/10.1109/TIM.2020.2984820>.

Xu, H., Guerrier, S., Molinari, R., & Karemera, M., “*Multivariate signal modeling with applications to inertial sensor calibration*”, IEEE Transactions on Signal Processing, 67(19), 5143-5152, 2019.
Full text: <https://doi.org/10.1109/TSP.2019.2935902>.

Branca, M., Orso, S., Molinari, R., **Xu, H.**, Guerrier, S., Zhang, Y., & Mili, N., “*Is nonmetastatic cutaneous melanoma predictable through genomic biomarkers?*”, Melanoma Research, 28(1), 21-29, 2018.
Full text: <https://doi.org/10.1097/CMR.0000000000000412>.

Zhang, Y., **Xu, H.**, Radi, A., Molinari, R., Guerrier, S., Karemera, M., & El-Sheimy, N., “*An optimal virtual inertial sensor framework using wavelet cross covariance*”, In 2018 IEEE/ION Position, Location and Navigation Symposium (PLANS) (1342-1350).
Full text: <https://doi.org/10.1109/PLANS.2018.8373525>.

Xu, H., Guerrier, S., Molinari, R., & Zhang, Y., “*A study of the Allan variance for constant-mean non-stationary processes*”, IEEE Signal Processing Letters, 24(8), 1257-1260, 2017.
Full text: <https://doi.org/10.1109/LSP.2017.2722222>

Preprints

Kumar, S., **Xu, H.**, Padilla, C.M.M., Khoo Y., Padilla, O.H.M., & Wang, D., “*Bias-variance Tradeoff in Tensor Estimation*”. (submitted).
Full text: <https://arxiv.org/abs/2509.17382>.

Xu, H., Padilla, C.M.M., Padilla, O.H.M., & Wang, D., “*Multivariate Poisson intensity estimation via low-rank tensor decomposition*”. (submitted).
Full text: <https://arxiv.org/abs/2504.15879>.

Voirol, L., **Xu, H.**, Zhang, Y., Insolia, L., Molinari, R., & Guerrier, S., “*Inference for large scale regression models with dependent errors*”. (preparing for submission).
Full text: <https://arxiv.org/abs/2409.05160>.

Xiao, D., **Xu, H.**, Ahn, J., Guerrier, S., Li, R., & Ke, Y., “*Multiple change points detection problems for*

high-dimensional time series". (submitted).

Xu, H., Guerrier, S., Li, R., & Ke, Y., "On tail-robust autocovariance matrix estimation for high-dimensional and potentially nonstationary time series", 2024.

Full text: <https://arxiv.org/abs/2609.04418>.

Ebooks

Guerrier, S., Molinari, R., **Xu, H.** & Zhang, Y., "Applied time series analysis with R".

Full text: <https://smac-group.github.io/ts/>.

Statistical Softwares

FragmentCP - R package: performs change point localisation and inference in fragmented functional data.

changepoints - R package: performs a series of offline and/or online change-point detection algorithms for numerous settings. Available on CRAN.

rcov - R package: collection of tools for estimating robust autocovariance matrix for high-dimensional time series.

avar - R package: implements the allan variance and allan variance linear regression estimator for time series models. Available on CRAN.

Grants

Swiss National Science Foundation (SNSF) [Postdoc.Mobility Fellowship](#) (USD 120,000), 2022-2023.

Travel Funding, Société Académique de Genève (USD 1400), 2018.

Invited Talks

"Bias-variance tradeoff in tensor decomposition", 2025 IMS International Conference on Statistics and Data Science (ICSIDS), Seville, Spain 12/2025.

"Multivariate Poisson intensity estimation via low-rank tensor decomposition", 2024 IMS International Conference on Statistics and Data Science (ICSIDS), Nice, France, 12/2024.

"Change point localisation and inference in high-dimensional regression models under dependence", Statistics Seminar, Fudan University, China, 08/2023.

"Online network change point detection with missing values and temporal dependence", Workshop on Change point Analysis, University of Warwick, UK, 05/2023.

"Online network change point detection with missing values", StatScale Seminar, online, 02/2023.

"Change point localisation and inference in high-dimensional regression models under dependence", SMAC Talk, Penn State University, USA, 04/2023.

"Change point localisation and inference in high-dimensional regression models under dependence", Statistics Seminar, University of Notre Dame, USA, 11/2022.

"Change point localisation and inference in high-dimensional regression models under dependence", ICMS workshop: Structural Breaks and Shape Constraints, Edinburgh, 05/2022.

"Robust Estimation of Large Autocovariance Matrices", 2021 ICSA Applied Statistics Symposium, online, 09/2021.

"Robust Estimation of Large Autocovariance Matrices", Statistics seminar, Université catholique de Louvain, 05/2021.

“Long-run Covariance Matrix Estimator for High-dimensional Time Series”, The 3rd International Conference on Econometrics and Statistics, National Chung Hsing University, Taiwan, 06/2019.

“A GMWM-based Inference for Correlated Latent Processes”, 2017 IMS-China International Conference on Statistics and Probability, Guangxi University For Nationalities, China, 06/2017.

Academic Visits Visiting student at University of Illinois at Urbana-Champaign, Feb–Jun 2016, Feb-May 2017
Visiting student at Penn State University, Feb–Jun 2018

Teaching experience **Instructor:**

- Categorical Data Analysis (graduate), Auburn University, Spring 2026

- Regression Analysis (graduate), Auburn University, Fall 2025-2026

- Bayesian forecasting and intervention (undergraduate/graduate), University of Warwick, Spring 2024 and 2025

- Elementary Mathematical Statistics (undergraduate), Penn State University, Spring 2023

Teaching Assistant: responsible for giving weekly recitation lectures/office hours, exam preparation and grading.

- Statistical Modeling (undergraduate), University of Geneva, Fall 2015-2020

- Business Analytics (undergraduate), University of Geneva, Fall 2016-2017

- Numerical Methods (undergraduate), University of Geneva, Fall 2020

- Statistics I (undergraduate), University of Geneva, Fall 2015-2020

- Mixed Linear Models (graduate), University of Geneva, Fall 2016-2019

Students Mentoring **Gengyu Xue**, PhD student at University of Warwick, 2023-2025. (co-advised with Prof. Yi Yu; ICSDS 2024 Student Travel Award).

Academic Services - Organizer of the Statistics & Data Science Seminar at Auburn University (2025-2026).
- Organizer of the Department of Statistics Research Seminar at University of Warwick (2023-2024, 2024-2025).
- Organizer of the invited session *New advances in statistical learning and simulation-based inference*, CMStatistics, Berlin, 2023.

Skills **Languages:** Chinese (native); English (fluent); French (elementary).

Computer Programming and Statistical Software: C++, R, SAS, Matlab, Python

Professional Experience **Statistician**, IMS Health (now IQVIA), Beijing, China, 10/2013–01/2014: Design statistical methods to investigate the causes of changes in trend of Rx data in mail order, retail order and longtime-care order. Programmed SAS, SQL and JCL code to manipulate Rx data and generate reproducible report.